Qubit Introduction

Martina Paola Calvi

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1 Introduction to Qubits

1.1 Computers and Computation

Models of Classical Computation

Classical computation can be implemented through various physical models:

- **Mechanical**: Curta calculator, Digi-comp I, Babbage's difference engine
- **Electrical**: Electronic circuits and transistor-based systems
- **Optical**: Photonic computing systems
- **Biological**: DNA and molecular computing

Conceptual Models

Key theoretical frameworks for computation include:

- Turing Machine (mathematical model of computation)
- Cellular Automata
- Von Neumann architecture (CPU, Memory, Bus, I/O)
- Logic-in-Memory computing paradigm

Quantum Computing Development

In the early 1980s, Richard Feynman proposed that simulating quantum systems requires quantum computers. This led to the fundamental question: what algorithms could provide a quantum advantage for real-world problems?

Two major milestones occurred in the mid-1990s:

1. Discovery of Shor's algorithm for factoring (addressing an important practical problem)
2. Development of quantum error-correcting codes by Shor and colleagues

Since then, research has focused on both the underlying physics and hardware development for quantum computers. Today, operating quantum computers with more than 100 qubits exist, with IBM's Condor exceeding 1000 qubits.

1.2 Data Representation: Classical vs. Quantum

Classical computers use bits (0 or 1), while quantum computers use quantum bits (qubits) that can exist in superposition: $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$ where $|\alpha|^2 + |\beta|^2 = 1$.

Quantum Parallelism

Unlike classical bits, quantum systems exhibit superposition, allowing exponential encoding of information. While N classical bits represent a single N-bit state, N qubits exist in a superposition of all 2^N possible states simultaneously.

For example, with $N = 3$ qubits:

$$|\psi\rangle = \sum_{i=0}^{7} c_i |i\rangle$$

where each c_i is a complex amplitude and $\sum_{i=0}^{7} |c_i|^2 = 1$.

This quantum parallelism enables quantum algorithms to process all 2^N computational paths concurrently, providing potential exponential speedup over classical approaches. However, measurement collapses this superposition to a single state, requiring careful algorithm design to extract useful information.

1.3 DiVincenzo Criteria for Quantum Computing

A viable qubit must satisfy the DiVincenzo Criteria:

1. Scalability of the physical system
2. Ability to initialize qubit states
3. Long coherence times (T_1, T_2, T_ϕ)
4. Universal quantum gate operations
5. High-fidelity quantum operations
6. Interconversion between stationary and flying qubits
7. Faithful transmission of flying qubits

1.4 Qubit Figure of Merits

Qubit Robustness and Coherence Time

Qubits lose information through interaction with the environment via:

- Energy relaxation (T_1): loss of energy to the environment
- Dephasing (T_ϕ): loss of phase coherence

The effective coherence time is $\frac{1}{T_2} = \frac{1}{T_\phi} + \frac{1}{2T_1}$, representing the average time a qubit remains coherent.

Gate Speed and Figure of Merit

The number of gates performable within qubit lifetime determines quantum advantage: $N_{\text{gates}} \sim t_{\text{gate}}/T_2$.

Gate Fidelity

Gate fidelity quantifies how accurately quantum gate operations are performed, measuring the closeness between ideal and actual operations under imperfections and noise. High-fidelity operations are crucial for reliable quantum computation and can be determined through Process Tomography or Randomized Benchmarking.

1.5 Bits vs. Qubits

Classical vs. Quantum Gates

Classical computers use Boolean logic gates (such as NOT and AND) to form a universal set of operations. Quantum computers similarly use quantum gates, but with fundamentally different properties:

- Classical gates: irreversible, map multiple inputs to outputs
- Quantum gates: unitary and reversible, preserve quantum information
- Single quantum gates (e.g., X-gate) can manipulate qubit states
- Universal quantum computation requires a combination of single and two-qubit gates

The X-gate example shows practical application: applying an X-gate to a qubit in state $|0\rangle$ transitions it to $|1\rangle$, achieved experimentally through a π-pulse envelope.

Circuits in Space vs. Circuits in Time

Classical logic operates as **circuits in space**:

- Example: NOT gate maps $0 \to 1$ and $1 \to 0$
- Horizontal axis represents space
- Input and output exist at different physical locations
- Measurements can be performed simultaneously on different wires

In contrast, quantum logic operates as **circuits in time**:

- Example: X gate (quantum NOT) acts on qubit states $|0\rangle$ and $|1\rangle$
- Horizontal axis represents time
- Input and output correspond to the same qubit before and after the operation
- State is sequentially updated through time evolution

Key distinction:

- Classical gates connect wires in space, allowing parallel independent operations
- Quantum gates are unitary operations that evolve the state of qubits in time
- Each gate application modifies and overwrites the qubit's state

Two-Qubit Gates

Two-qubit gates enable quantum entanglement and are essential for universal quantum computation.

CNOT Gate (Controlled-NOT): The CNOT gate operates on two qubits: a control qubit and a target qubit. It applies an X gate to the target if the control qubit is $|1\rangle$.

$$\text{CNOT} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}$$

Truth table:

Control	Target	Control	Target
$\|0\rangle$	$\|0\rangle$	$\|0\rangle$	$\|0\rangle$
$\|0\rangle$	$\|1\rangle$	$\|0\rangle$	$\|1\rangle$
$\|1\rangle$	$\|0\rangle$	$\|1\rangle$	$\|1\rangle$
$\|1\rangle$	$\|1\rangle$	$\|1\rangle$	$\|0\rangle$

For a control qubit in superposition $|\psi\rangle_A = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$ and target in state $|0\rangle_B$:

$$|\psi\rangle_{\text{out}} = \text{CNOT}(|\psi\rangle_A \otimes |0\rangle_B) = \frac{1}{\sqrt{2}}(|0\rangle_A|0\rangle_B + |1\rangle_A|1\rangle_B)$$

This creates an entangled Bell state, where the qubits are correlated regardless of measurement basis.

1.6 Qubit States

General Qubit State

Quantum mechanics tells us that any qubit system can exist in a superposition of states. The general state of a quantum bit is described by:

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

where α and β are complex numbers, and the normalization constraint $\langle\psi|\psi\rangle = 1$ requires that:

$$|\alpha|^2 + |\beta|^2 = 1$$

The states $|0\rangle$ and $|1\rangle$ represent the computational basis as two-dimensional vectors:

$$|0\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad |1\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

A general superposition state can be expressed as a weighted sum of basis states:

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle = \alpha \begin{pmatrix} 1 \\ 0 \end{pmatrix} + \beta \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} \alpha \\ \beta \end{pmatrix}$$

Bra-Ket Notation and Inner Products

The transpose complex conjugate of a ket $|\psi\rangle$ is denoted as a bra:

$$|\psi\rangle^\dagger = \langle\psi| = \begin{pmatrix}\alpha^* & \beta^*\end{pmatrix}$$

where the † superscript indicates the Hermitian conjugate (transpose complex conjugate). The inner product forms the bra-ket:

$$\langle\psi|\psi\rangle = |\alpha|^2 + |\beta|^2 = 1$$

This represents the normalization condition ensuring the total probability is conserved.

Measurement of Qubits

When a qubit in superposition $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$ is measured in the computational basis:

- The measurement is an active process where the apparatus interacts with the qubit
- The qubit state collapses to either $|0\rangle$ or $|1\rangle$
- The probability of measuring $|0\rangle$ is $P(0) = |\alpha|^2$
- The probability of measuring $|1\rangle$ is $P(1) = |\beta|^2$
- After measurement, the qubit is no longer in superposition
- Only partial information is extracted—you cannot access the full quantum state
- Information obtained depends on the chosen measurement basis

This measurement in the computational basis $\{|0\rangle, |1\rangle\}$ is called **Z-measurement**. Other measurement bases exist (X-basis, Y-basis), but the computational basis is most common in quantum computing systems.

The Born Rule

The probability of measuring state $|a\rangle$ from $|\psi\rangle$ is:

$$P(a) = |\langle a|\psi\rangle|^2$$

Common measurement bases include:

- Computational (Z-basis): $\{|0\rangle, |1\rangle\}$
- X-basis: $\{|+\rangle, |-\rangle\}$ where $|+\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$
- Y-basis: $\{|+i\rangle, |-i\rangle\}$ where $|+i\rangle = \frac{1}{\sqrt{2}}(|0\rangle + i|1\rangle)$

TODO:Bloch Sphere Representation

Any single-qubit pure state can be represented on the Bloch sphere:

$$|\psi\rangle = \cos\frac{\theta}{2}|0\rangle + e^{i\phi}\sin\frac{\theta}{2}|1\rangle$$

where $\theta \in [0, \pi]$ and $\phi \in [0, 2\pi]$.
The Bloch vector is $\vec{r} = (\sin\theta\cos\phi, \sin\theta\sin\phi, \cos\theta)$.
Key states: $|0\rangle \to (0, 0, 1)$, $|1\rangle \to (0, 0, -1)$, $|+\rangle \to (1, 0, 0)$, $|-\rangle \to (-1, 0, 0)$.

1.7 Single-Qubit Gates

Single-qubit gates are unitary transformations that evolve qubit states. A unitary matrix U satisfies $U^\dagger U = UU^\dagger = I$.

Pauli Gates

Pauli-X (NOT gate):

$$\sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma_x|0\rangle = |1\rangle, \quad \sigma_x|1\rangle = |0\rangle$$

Represents a π rotation around the x-axis.

Pauli-Z gate:

$$\sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \quad \sigma_z|0\rangle = |0\rangle, \quad \sigma_z|1\rangle = -|1\rangle$$

Represents a π rotation around the z-axis.

Pauli-Y gate:

$$\sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} = -i\sigma_x\sigma_z$$

Represents a π rotation around the y-axis. All Pauli matrices satisfy $\sigma_i^2 = I$ and are Hermitian.

Commutation Relations

$$[\sigma_x, \sigma_y] = 2i\sigma_z, \quad [\sigma_y, \sigma_z] = 2i\sigma_x, \quad [\sigma_z, \sigma_x] = 2i\sigma_y$$

Anticommutation Relations

$$\{\sigma_x, \sigma_y\} = 0, \quad \{\sigma_y, \sigma_z\} = 0, \quad \{\sigma_z, \sigma_x\} = 0$$

Rotation Gates

General rotation around axis n by angle θ:

$$R_n(\theta) = e^{-i\theta\sigma_n/2} = \cos\frac{\theta}{2}I - i\sin\frac{\theta}{2}\sigma_n$$

Hadamard Gate

$$H = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$$

Creates superposition: $H|0\rangle = |+\rangle$, $H|1\rangle = |-\rangle$.

Phase Gate (S-gate)

$$S = \begin{pmatrix} 1 & 0 \\ 0 & i \end{pmatrix}$$

Adds a 90 phase shift: $S|+\rangle = |+i\rangle$, $S|-\rangle = |-i\rangle$.

1.8 Multiple Gate Applications

Sequential gate applications are computed via matrix multiplication (right to left in mathematical notation, left to right in circuits).

1.9 Density Matrix Formalism

Real quantum systems cannot be perfectly isolated and are described by the density matrix $\hat{\rho}$:

Pure States

$$\hat{\rho} = |\psi\rangle\langle\psi|$$

Mixed States (Ensemble)

$$\hat{\rho} = \sum_i p_i |\psi_i\rangle\langle\psi_i|$$

where $\sum_i p_i = 1$.

Properties

- Hermitian: $\hat{\rho} = \hat{\rho}^\dagger$
- Trace normalized: $\mathrm{tr}(\hat{\rho}) = 1$
- 2×2 matrix form: $\hat{\rho} = \begin{pmatrix} \rho_{11} & \rho_{12} \\ \rho_{21} & \rho_{22} \end{pmatrix}$
- Diagonal elements represent probabilities
- Off-diagonal elements represent coherences (with $\rho_{12} = \rho_{21}^*$)

Pure vs. mixed states can be distinguished by $\mathrm{tr}(\hat{\rho}^2) = 1$ (pure) or $\mathrm{tr}(\hat{\rho}^2) < 1$ (mixed).

2 Superconducting Qubits

Superconductivity Review

In 1911, Heike Kamerlingh Onnes discovered superconductivity by observing that the resistivity of mercury vanished abruptly at 4.2 K. In 1933, Walther Meissner and Robert Ochsenfeld discovered that below a critical temperature T_C, a superconductor expels magnetic fields from its interior—the **Meissner effect**—demonstrating that superconductivity is a distinct thermodynamic state, not merely a perfect conductor.

In 1957, John Bardeen, Leon Cooper, and J. Robert Schrieffer published the microscopic **BCS Theory** of superconductivity. The key insight is that electrons form **Cooper pairs**—correlated pairs with opposite momentum and spin. These pairs form a coherent **Bose-Einstein condensate** described by a single macroscopic wave function:

$$\psi(\mathbf{r}, t) = \sqrt{\rho_s(\mathbf{r}, t)} e^{i\phi(\mathbf{r},t)}$$

where ρ_s is the density of Cooper pairs and ϕ is the macroscopic phase. The superconducting current is related to the phase gradient:

$$J_s = \frac{n_s e\hbar}{m} \nabla\phi - \frac{n_s e^2}{mc} \mathbf{A}$$

This phase rigidity—where variations in ϕ cost energy—is the microscopic origin of superconductivity.

The Josephson Junction

In 1962, Brian D. Josephson analyzed a junction of two superconductors separated by a thin insulating barrier (0.1–1 nm). Despite the barrier, Cooper pairs tunnel across via the quantum tunneling effect. The junction is characterized by two fundamental relations:

DC (Current-Phase) Relation:

$$I = I_C \sin\phi$$

where I_C is the critical current (maximum supercurrent without dissipation) and $\phi = \phi_2 - \phi_1$ is the phase difference.

AC (Voltage-Phase) Relation:

$$V = \frac{\Phi_0}{2\pi} \frac{d\phi}{dt}$$

where $\Phi_0 = h/2e = 2\pi\hbar/2e$ is the **magnetic flux quantum**. This shows that voltage drives phase evolution in time.

Josephson Inductance and Energy

The junction exhibits non-linear inductance. Differentiating the current-phase relation:

$$\frac{dI}{dt} = I_C \cos(\phi) \frac{d\phi}{dt}$$

Substituting $\frac{d\phi}{dt} = \frac{2\pi V}{\Phi_0}$ yields:

$$L_J(\phi) = \frac{\Phi_0}{2\pi I_C \cos(\phi)} = \frac{L_{J0}}{\cos(\phi)}, \quad L_{J0} = \frac{\Phi_0}{2\pi I_C}$$

The Josephson energy (potential energy stored in phase) is:

$$E_J(\phi) = -E_J \cos\phi$$

where $E_J = \frac{\Phi_0 I_C}{2\pi} = \frac{\hbar I_C}{2e}$ is the **Josephson coupling energy**.

Energy Scales: E_J and E_C

Superconducting qubits are characterized by two competing energy scales:

- **Josephson Energy (E_J):** Tunneling energy of Cooper pairs across the barrier.
- **Charging Energy (E_C):** Energy cost to add one Cooper pair to the island: $E_C = \frac{e^2}{2C}$.

The ratio E_J/E_C determines the qubit regime and dynamical properties.

Circuit Quantization

Using canonical quantization, flux Φ and charge Q become non-commuting operators:

$$[\hat{\Phi}, \hat{Q}] = i\hbar$$

The Hamiltonian for a superconducting qubit with a Josephson junction replacing the inductor is:

$$\hat{H} = \frac{\hat{Q}^2}{2C} - E_J \cos\left(\frac{2\pi\hat{\Phi}}{\Phi_0}\right)$$

In terms of the phase operator $\hat{\phi} = 2\pi\hat{\Phi}/\Phi_0$ and the Cooper pair number operator $\hat{n} = \hat{Q}/2e$:

$$\hat{H} = 4E_C(\hat{n} - n_g)^2 - E_J \cos\hat{\phi}$$

where n_g is the gate charge offset. The term $(\hat{n} - n_g)^2$ means the qubit is sensitive to charge noise from the environment.

The Transmon Regime

The **Transmon** (Transmission-line shunted plasma oscillation qubit), introduced by Koch et al. (2007), is the industry standard. It operates in the regime $E_J/E_C \gg 1$ (typically 20–100) to minimize charge noise sensitivity.

Exponential Noise Suppression: By increasing E_J/E_C, energy level dispersions (charge noise sensitivity) are exponentially suppressed:

$$\text{Dispersion} \propto \exp\left(-\sqrt{\frac{8E_J}{E_C}}\right)$$

For $E_J/E_C = 50$: $\exp(-\sqrt{400}) \approx \exp(-20) \approx 10^{-9}$. This provides dephasing times $T_2 > 100\ \mu$s.

Controlled Anharmonicity: Despite increased immunity to noise, the Transmon retains small anharmonicity. Expanding the cosine potential:

$$-E_J \cos\hat{\phi} \approx E_J \left(\frac{\hat{\phi}^2}{2} - \frac{\hat{\phi}^4}{24} + \cdots \right)$$

In terms of ladder operators and applying the Rotating Wave Approximation:

$$\hat{H} \approx \hbar\omega_T \hat{b}^\dagger \hat{b} - \frac{E_C}{12}(\hat{b} + \hat{b}^\dagger)^4$$

where $\omega_T = \sqrt{8E_J E_C}/\hbar - E_C/\hbar$ is the transition frequency. The anharmonicity is:

$$\alpha = E_{1\to 2} - E_{0\to 1} = -E_C$$

Typical values: $E_J \sim 20$ GHz, $E_C \sim 300$–400 MHz, yielding anharmonicity of 6–8

Typical Performance Metrics:

- Relaxation time: $T_1 \sim 200$–500 μs
- Dephasing time: $T_2 \sim 100$–300 μs
- Gate fidelity: $> 99\%$
- Readout fidelity: $> 95\%$

Other Superconducting Qubit Types

Beyond the Transmon, alternative architectures include:

- **Cooper Pair Box (CPB):** The original charge qubit (Nakamura et al., Nature 1999), operating in the regime $E_J < E_C$. Highly sensitive to charge noise; historically important but superseded by Transmon.
- **Flux Qubit:** A superconducting loop with 3 Josephson junctions, operated near half a flux quantum ($\Phi_{\text{ext}} \approx \Phi_0/2$). The two persistent current states form the qubit basis. Sensitive to flux noise but with intermediate coherence times.
- **Fluxonium:** An island connected to ground through a superinductance (many large junctions in parallel) with a small junction providing strong non-linearity. High anharmonicity ($\alpha \sim 1$ GHz), robust against both charge and flux noise. Coherence times $T_2 > 1$ ms have been demonstrated.
- **Phase Qubit:** Exploits energy quantization in a double-well potential formed by a junction operated near critical current. Historically important (UCSB, 2002–2007) but coherence times are limited.

Decoherence Mechanisms and Coherence Times

Decoherence is characterized by two fundamental timescales:

- T_1 **(Relaxation):** Energy decay from $|1\rangle$ to $|0\rangle$. Sources: dielectric losses in the substrate, dielectric loss at the tunnel junction, Purcell damping from environmental modes, quasiparticle poisoning. State-of-the-art Transmons achieve $T_1 \sim 200$–500 μs.
- T_2 **(Dephasing):** Coherence decay. Related to T_1 by $T_2^{-1} = (2T_1)^{-1} + T_\phi^{-1}$, where T_ϕ is pure dephasing caused by 1/f flux noise, surface charge defects, and other low-frequency noise. Best Transmons achieve $T_2 \sim 100$–300 μs.

3 Circuit QED

3.1 From Atomic Physics to Quantum Optics

The foundation of Circuit QED lies in coherent light-matter interaction at the quantum level. In atomic physics, discrete energy levels allow photons to drive transitions between states. The simplest useful model reduces an atom to a two-level system (TLS) with ground state $|0\rangle$ and excited state $|1\rangle$.

The Two-Level System

The transition energy is:

$$E_{01} = E_1 - E_0 = \hbar\omega_{01}$$

Coupling to a classical oscillating electric field is described by:

$$H = -\vec{d} \cdot \vec{E}(t), \quad E(t) = E_0 \cos(\omega_{01} t)$$

where d is the transition dipole moment. Near-resonant driving produces Rabi oscillations—coherent energy exchange between atom and field.

Vacuum Fluctuations and Field Confinement

A key question motivates Cavity QED: can vacuum fluctuations alone produce significant effects on a qubit? The vacuum electric field amplitude in a single-mode cavity is:

$$E_{\text{zpf}} = \sqrt{\frac{\hbar\omega_r}{2\varepsilon_0 V}}$$

In optical cavities, $V \sim \lambda^3$ gives $E_{\text{zpf}} \sim 10^{-4}$ V/m. Superconducting circuits achieve $V_{\text{eff}} \sim 10^{-3}\lambda^3$, yielding $E_{\text{zpf}} \approx 0.2$ V/m—orders of magnitude larger. This tight confinement enables single-photon effects.

3.2 Cavity QED and the Fabry-Pérot Resonator

Traditional Cavity QED places atoms inside Fabry-Pérot cavities—two highly reflective mirrors separated by distance L. Resonant modes satisfy:

$$2L = \lambda_m m, \quad f_m = \frac{mc}{2L}$$

where m is the mode number. The quality factor quantifies resonator performance:

$$Q = \frac{f_{\text{resonance}}}{\Delta f_{\text{3dB}}} = \frac{\omega_r}{\kappa}$$

Superconducting microwave resonators achieve $Q \sim 10^4$–10^6, corresponding to photon lifetimes $1/\kappa$ of microseconds.

The vacuum Rabi coupling constant is:

$$g = \frac{d \cdot E_{\text{zpf}}}{\hbar}$$

In optical cavity QED: $g/2\pi \sim 50$ kHz. In circuit QED: $g/2\pi \sim 0$–1 GHz—five orders of magnitude stronger due to large artificial dipole moments and tight field confinement.

3.3 Circuit QED: Artificial Atoms and Coplanar Waveguides

Circuit QED transplants cavity QED physics to a microchip, replacing atoms with superconducting qubits and Fabry-Pérot cavities with coplanar waveguide (CPW) resonators.

Advantages of Superconducting Qubits

Superconducting qubits possess effective dipole moments several hundred to thousand times larger than atoms because charge is distributed over micrometer-scale distances:

- Large engineered dipole moment $\rightarrow$ strong coupling g (hundreds of MHz)
- On-chip frequencies determined by design (lithography), not fixed by nature
- Integrated fabrication $\rightarrow$ scalable multi-qubit processors
- Fast gate operations (nanosecond timescale) enabled by large g

Coplanar Waveguide Resonators

The CPW resonator consists of a superconducting transmission line (Nb or Al on sapphire) with center conductor separated from ground planes. At microwave frequencies (3–8 GHz), the resonator length is $L \sim \lambda/2$. The electromagnetic field is confined to volume:

$$V_{\text{eff}} \sim w \cdot d \cdot L \sim 10^{-3}\lambda^3$$

where w and d are conductor width and gap (sub-micron). This yields zero-point voltage fluctuations:

$$V_{\text{zpf}} = \sqrt{\frac{\hbar\omega_r}{2C_r}} \approx 1\ \mu\text{V}$$

The enormous field concentration enables coupling rates in the hundreds of MHz range.

3.4 Resonator Quantization and One-Mode Approximation

Quantum LC Oscillator

A CPW section is modeled as a lumped LC circuit with resonance frequency:

$$\omega_r \approx \frac{1}{\sqrt{L_r C_r}}$$

Quantizing via canonical commutation relations $[\hat{Q}, \hat{\Phi}] = i\hbar$ and introducing ladder operators:

$$\hat{\Phi} = \Phi_{\text{zpf}}(\hat{a} + \hat{a}^\dagger), \quad \hat{Q} = iQ_{\text{zpf}}(\hat{a}^\dagger - \hat{a})$$

where $\Phi_{\text{zpf}} = \sqrt{\hbar/(2\omega_r C_r)}$, the Hamiltonian becomes:

$$\hat{H}_r = \hbar\omega_r \hat{a}^\dagger \hat{a}$$

One-Mode Approximation

A real transmission line supports infinitely many modes at frequencies $\omega_1, \omega_2, \omega_3, \ldots$ with spacing of several GHz. The linewidth $\kappa_m \sim$ MHz is much smaller than mode spacing:

$$\kappa_m \ll |\omega_{m+1} - \omega_m|$$

Near resonance, the qubit couples appreciably to only one mode ω_r. All others decouple, justifying the single-mode approximation.

3.5 The Full Circuit QED Hamiltonian

The total system Hamiltonian combines resonator, qubit, and interaction. Using the two-level approximation for the transmon and ladder operators:

$$\hat{H} = \hbar\omega_r \hat{a}^\dagger \hat{a} + \frac{\hbar\omega_{01}}{2}\hat{\sigma}_z - g(\hat{\sigma}_+ - \hat{\sigma}_-)(\hat{a}^\dagger - \hat{a})$$

where $\hat{a}$ ($\hat{\sigma}_\pm$) are resonator (qubit) operators, and the coupling strength is:

$$g = \omega_r \frac{C_g}{C}\left(\frac{E_J}{2E_C}\right)^{1/4}\sqrt{\frac{\pi Z_r}{R_k}}$$

with $Z_r \approx 50\ \Omega$ (resonator impedance) and $R_k = h/e^2 \approx 25.8$ kΩ (von Klitzing constant).

3.6 The Jaynes-Cummings Model

The Jaynes-Cummings (JC) Hamiltonian is the cornerstone of cavity and circuit QED. It emerges from the full Hamiltonian via the Rotating Wave Approximation.

Rotating Wave Approximation

Expanding the interaction term yields four products:

$$-g(\hat{\sigma}_+\hat{a}^\dagger - \hat{\sigma}_+\hat{a} - \hat{\sigma}_-\hat{a}^\dagger + \hat{\sigma}_-\hat{a})$$

The terms $\hat{\sigma}_+\hat{a}^\dagger$ and $\hat{\sigma}_-\hat{a}$ represent energy non-conserving processes oscillating at $\omega_r + \omega_{01}$. Near resonance ($\omega_r \approx \omega_{01}$), these oscillate rapidly and can be dropped. Energy-conserving terms $\hat{\sigma}_+\hat{a}$ and $\hat{\sigma}_-\hat{a}^\dagger$ evolve slowly.

After RWA, the Jaynes-Cummings Hamiltonian is:

$$\hat{H}_{\text{JC}} = \hbar\omega_r \hat{a}^\dagger \hat{a} + \frac{\hbar\omega_{01}}{2}\hat{\sigma}_z + \hbar g(\hat{a}^\dagger\hat{\sigma}_- + \hat{a}\hat{\sigma}_+)$$

The interaction terms enable coherent photon-qubit exchange while conserving energy.

Dressed States and Vacuum Rabi Splitting

Exactly at resonance ($\Delta = \omega_{01} - \omega_r = 0$), the JC model yields energy eigenstates:

$$|+\rangle = \frac{|1,0\rangle + |0,1\rangle}{\sqrt{2}}, \quad |-\rangle = \frac{|1,0\rangle - |0,1\rangle}{\sqrt{2}}$$

with energies $E_\pm = \hbar\omega_r \pm \hbar g$. The energy splitting $2\hbar g$ at resonance is the vacuum Rabi splitting, observable in transmission spectroscopy as two peaks—direct evidence of strong light-matter coupling.

3.7 Qubit Readout: The Dispersive Regime

Resonant coupling enables fast energy exchange (useful for gates) but is destructive for non-demolition measurement. Readout uses the dispersive regime.

Dispersive Hamiltonian

When the qubit is far detuned from the resonator ($\Delta = \omega_{01} - \omega_r \gg g, \kappa$), second-order perturbation theory yields:

$$\hat{H}_{\text{disp}} \approx \hbar(\omega_r + \chi\hat{\sigma}_z)\hat{a}^\dagger\hat{a} + \frac{\hbar\omega_{01}}{2}\hat{\sigma}_z$$

where the dispersive shift is:

$$\chi = \frac{g^2}{\Delta}$$

The resonator frequency becomes qubit-state-dependent:

- Qubit state $|0\rangle \to \omega_r - \chi$
- Qubit state $|1\rangle \to \omega_r + \chi$

Quantum Non-Demolition Readout

By sending a probe tone at frequency $\omega_{\text{probe}} \approx \omega_r$, the reflected (or transmitted) signal acquires a state-dependent phase shift:

$$\phi(|0\rangle) = -\arctan\left(\frac{2(\omega_{\text{probe}} - \omega_r + \chi)}{\kappa}\right)$$

$$\phi(|1\rangle) = -\arctan\left(\frac{2(\omega_{\text{probe}} - \omega_r - \chi)}{\kappa}\right)$$

Measuring $\Delta\phi = \phi(|0\rangle) - \phi(|1\rangle)$ determines the qubit state without exchanging energy. Readout fidelity is limited only by noise and integration time.

Typical Circuit QED Parameters

- Coupling: $g/2\pi \approx 50$–300 MHz
- Detuning: $\Delta/2\pi \approx 0.5$–3 GHz (ensuring $\Delta \gg g$)
- Dispersive shift: $\chi/2\pi = g^2/\Delta \approx 0.5$–$10$ MHz
- 2D resonator linewidth: $\kappa/2\pi \approx 5$–10 MHz
- 3D cavity linewidth: $\kappa/2\pi \approx 1$–10 kHz

3.8 Single-Qubit Control

XY Control (Rabi Driving)

Microwave pulses at frequency $\omega_d \approx \omega_{01}$ drive Rabi oscillations. In the rotating frame (frame rotating at ω_d), the effective Hamiltonian is:

$$\hat{H}_{\text{drive}} = \frac{\hbar\Delta_d}{2}\hat{\sigma}_z + \frac{\hbar\Omega_R}{2}(\cos\phi\hat{\sigma}_x + \sin\phi\hat{\sigma}_y)$$

where $\Delta_d = \omega_{01} - \omega_d$ is the drive detuning and $\Omega_R = d \cdot E_0/\hbar$ is the Rabi frequency controlled by pulse amplitude E_0. The phase ϕ determines the rotation axis (X or Y gate).

In the dispersive regime, control photons are off-resonance with the cavity and mostly reflected. Careful pulse shaping (DRAG—Derivative Removal via Adiabatic Gate) suppresses unwanted qubit state transitions.

Z Control (Frequency Tuning)

Modern transmons incorporate a SQUID loop (Superconducting Quantum Interference Device) replacing the single Josephson junction. The effective Josephson energy becomes:

$$E_J(\Phi_{\text{ext}}) = E_{J,\max}\left|\cos\left(\pi\frac{\Phi_{\text{ext}}}{\Phi_0}\right)\right|$$

where Φ_{ext} is external magnetic flux through the SQUID and $\Phi_0 = h/(2e)$ is the flux quantum. Since $\omega_{01} \propto \sqrt{E_J}$, tuning Φ_{ext} via a local Z-control line enables fast (nanosecond) qubit frequency control.

3.9 The Transmon and Xmon Qubits

Transmon: Charge Insensitivity

The Cooper Pair Box has Hamiltonian:

$$H_{\text{CPB}} = 4E_C(\hat{n} - n_g)^2 - E_J\cos\hat{\phi}$$

where n_g is the gate charge offset. The CPB is highly charge-sensitive, making it susceptible to charge noise.

The transmon regime ($E_J/E_C \gg 1$, typically 50–100) is achieved by adding a large shunt capacitor. Results:

- Energy levels become nearly charge-insensitive (exponential suppression: $e^{-\sqrt{8E_J/E_C}}$)
- Qubit frequency: $\omega_{01} \approx \sqrt{8E_C E_J}/\hbar - E_C/\hbar$
- Anharmonicity: $\alpha = (E_{12} - E_{01})/\hbar \approx -E_C/\hbar$ (typically -100 to -300 MHz)
- Coherence times: $T_1 \sim 200$–500 μs, $T_2 \sim 100$–300 μs
- Gate fidelity: $> 99\%$

Xmon: Cross-Shaped Capacitor

The Xmon qubit replaces the transmon's capacitor with a cross-shaped ('X') metal pad. Advantages:

- Large shunt capacitance in compact footprint
- Simplified multi-qubit layout (four arms connect to readout, XY drive, Z control, quantum bus)
- Reduced parasitic coupling between qubits (geometric symmetry)
- Identical Hamiltonian to transmon
- Achieves $T_1, T_2 \sim$ 10s–100s of microseconds

Meander Resonators

Modern processors employ meandering (serpentine) CPW resonators, where the transmission line is folded to achieve resonance at shorter physical footprint. Benefits:

- Reduced chip area and crosstalk
- Simplified multi-qubit layout
- Same resonance frequency as straight resonators

3.10 Scalable Multi-Qubit Processors

Circuit QED provides a complete platform for scalable quantum processors:

Component	**Function**	**Implementation**
Readout Resonator	Measure qubit state	Meandered CPW
Xmon Island	Store quantum state	Al cross capacitor
SQUID/JJ	Non-linearity & tunability	$Al/Al_2O_3/Al$ junction
XY Control Line	Rotate qubits (gates)	Microwave drive line
Z Control Line	Tune qubit frequency	Local flux line
Quantum Bus	Entangle neighboring qubits	CPW coupler

These elements enable coherent control via resonant microwave drives and high-fidelity readout via dispersive interactions, forming the foundation of superconducting quantum processors with 10s to 100s of qubits.

4 Qubit Control

Recap on Readout

All superconducting qubits are currently measured primarily in the circuit quantum electrodynamics architecture. The circuit QED architecture involves a qubit coupled to a resonator, which in turn is coupled to a feed line, simulating the atom-cavity interaction of cavity electrodynamics.

In the dispersive regime, when the qubit is far detuned from the resonator frequency compared with their coupling rate g and the resonator linewidth κ (so $\Delta \gg g, \kappa$), the Jaynes-Cummings Hamiltonian can be approximated as:

$$H \approx (\omega_r + \chi\hat{\sigma}_z)\hat{a}^\dagger\hat{a} + \frac{\omega_{01}}{2}\hat{\sigma}_z$$

The resonator acquires a qubit-state-dependent shift. The hybridization of the qubit-resonator states results in decay of the qubit into the environment via the resonator's loss-rate κ, called **Purcell decay**:

$$\gamma_p = \frac{g^2}{\Delta^2}\kappa$$

This puts a constraint on how high κ can be. The ratio χ/κ is typically set to unity for optimal readout (balancing dispersive shift and cavity linewidth).

Purcell Filters

In some designs, **Purcell filters** are incorporated—frequency-selective elements placed between the resonator and transmission line. They act:

- Transparent at ω_r (measurement photons can leak)
- Opaque at ω_q (qubit-frequency photons are confined)

This design decouples measurement speed from qubit lifetime: $\kappa_{\text{eff}}(\omega_q) \ll \kappa(\omega_r)$.

4.1 Controlling the Qubit: The Bloch Sphere Picture

Reading out the qubit state is only half the task. To run quantum algorithms, we must also prepare arbitrary qubit states and perform single-qubit gates—unitary rotations of the state vector on the Bloch sphere.

For a transmon qubit, there are two physically distinct control mechanisms:

- **Electric (Capacitive / XY) Control:** A resonant microwave drive rotates the Bloch vector around an axis in the equatorial plane. The rotation axis is set by the phase of the drive, and the rotation angle by the pulse area.
- **Magnetic (Flux / Z) Control:** Tuning the qubit frequency via an external magnetic flux through the SQUID loop rotates the Bloch vector around the Z-axis.

4.2 Capacitive (XY) Control: Derivation

Circuit Model

A microwave drive voltage $V_d(t)$ is connected to the qubit island via a small coupling capacitor C_d (with $C_d \ll C_\Sigma$, where $C_\Sigma = C + C_d$ is the total qubit capacitance). The coupling is deliberately weak to preserve qubit coherence: a small C_d reduces the admittance seen from the drive line and limits energy leakage into the 50 Ω wiring impedance R_w.

The full circuit Hamiltonian (qubit plus drive coupling) is:

$$H = \frac{\hat{Q}^2}{2C_\Sigma} + \frac{\hat{\Phi}^2}{2L} + \frac{C_d}{C_\Sigma} V_d(t)\hat{Q}$$

The first two terms are the qubit oscillator. The third is the drive coupling: it is linear in the charge operator $\hat{Q}$ because the drive voltage couples electrostatically to the qubit island.

Quantizing the Charge Operator

Expressing the charge operator in terms of ladder operators (as done for the LC oscillator):

$$\hat{Q} = -iQ_{\text{zpf}}(\hat{a} - \hat{a}^\dagger), \quad Q_{\text{zpf}} = \sqrt{\frac{\hbar}{2Z_0}}, \quad Z_0 = \sqrt{\frac{L}{C_\Sigma}}$$

where Q_{zpf} is the charge zero-point fluctuation and Z_0 is the characteristic impedance of the qubit oscillator. Substituting into the drive coupling term:

$$H_{\text{drive}} = \frac{C_d}{C_\Sigma} V_d(t) \cdot (-iQ_{\text{zpf}})(\hat{a} - \hat{a}^\dagger)$$

Two-Level Truncation

For the transmon in the qubit subspace $\{|0\rangle, |1\rangle\}$, the ladder operators map to Pauli raising/lowering operators: $\hat{a} \to \hat{\sigma}_-$ and $\hat{a}^\dagger \to \hat{\sigma}_+$. The relevant Pauli matrix identities are:

$$\hat{\sigma}_+ = |1\rangle\langle 0|, \quad \hat{\sigma}_- = |0\rangle\langle 1|$$

$$\hat{\sigma}_x = \hat{\sigma}_+ + \hat{\sigma}_-, \quad \hat{\sigma}_y = i(\hat{\sigma}_- - \hat{\sigma}_+)$$

After truncation, $\hat{a} - \hat{a}^\dagger \to \hat{\sigma}_- - \hat{\sigma}_+ = i\hat{\sigma}_y$. The full Hamiltonian in the two-level approximation becomes:

$$\hat{H} = -\frac{\hbar\omega_q}{2}\hat{\sigma}_z + \Omega V_d(t)\hat{\sigma}_y$$

where $\omega_q = (E_1 - E_0)/\hbar$ is the qubit transition frequency and $\Omega = \frac{C_d}{C_\Sigma}\sqrt{\frac{\hbar}{2Z_0}}$ is the coupling strength (units: V^{-1}).

4.3 The Rotating Frame

The static qubit Hamiltonian $H_0 = -\frac{\hbar\omega_q}{2}\hat{\sigma}_z$ causes the Bloch vector to precess around the Z-axis at frequency $\omega_q \sim 5$ GHz. This rapid precession makes it difficult to visualize and analyze slow drive-induced rotations. The solution is to move into the rotating frame—a frame co-rotating with the qubit at ω_q.

Unitary Transformation

The rotating-frame transformation is defined by the unitary operator:

$$U(t) = e^{i\omega_q t\hat{\sigma}_z/2}$$

This represents a rotation of angle $\omega_q t$ around the Z-axis of the Bloch sphere. The new state in the rotating frame is:

$$|\psi_{\text{rf}}\rangle = U(t)|\psi\rangle$$

The effective Hamiltonian governing the evolution of $|\psi_{\text{rf}}\rangle$ is found via:

$$\hat{H}' = U(t)\hat{H}U^\dagger(t) - i\hbar U(t)\frac{dU^\dagger(t)}{dt}$$

The second term, $-i\hbar U(t)\frac{dU^\dagger}{dt} = -\frac{\hbar\omega_q}{2}\hat{\sigma}_z$, exactly cancels the free evolution term $U\hat{H}_0U^\dagger = -\frac{\hbar\omega_q}{2}\hat{\sigma}_z$. This is the defining property of the rotating frame: the free precession disappears and only the drive-induced dynamics remain.

Transforming the Drive Term

The drive term $\hat{H}_d = \Omega V_d(t)\hat{\sigma}_y$ must also be transformed. The key identity, derived from the Baker-Campbell-Hausdorff formula, is:

$$U(t)\hat{\sigma}_y U^\dagger(t) = \cos(\omega_q t)\hat{\sigma}_y - \sin(\omega_q t)\hat{\sigma}_x$$

Physically: $\hat{\sigma}_y$ in the lab frame becomes a combination of $\hat{\sigma}_x$ and $\hat{\sigma}_y$ in the rotating frame, rotating at ω_q. The drive Hamiltonian in the rotating frame is therefore:

$$\hat{H}'_d = \Omega V_d(t)[\cos(\omega_q t)\hat{\sigma}_y - \sin(\omega_q t)\hat{\sigma}_x]$$

4.4 I/Q Decomposition and the Rotating Wave Approximation

I/Q Signal Decomposition

A general monochromatic drive voltage at frequency ω_d with phase ϕ and envelope $s(t)$ can be written as:

$$V_d(t) = V_0 s(t)[\cos(\phi)\sin(\omega_d t) + \sin(\phi)\cos(\omega_d t)]$$

This is the standard I/Q (in-phase / quadrature) decomposition:

$$V_d(t) = V_0 s(t)[I\sin(\omega_d t) + Q\cos(\omega_d t)], \quad I = \cos(\phi), \quad Q = \sin(\phi)$$

Substituting the I/Q form into the rotating-frame drive Hamiltonian and expanding all products of trigonometric functions yields terms oscillating at the difference frequency $\delta\omega = \omega_q - \omega_d$ and the sum frequency $\omega_q + \omega_d$.

Rotating Wave Approximation (RWA)

When the drive is near resonance ($\omega_d \approx \omega_q$), the sum-frequency terms oscillate at $\sim 2\omega_q \sim 10$ GHz, which is much faster than the dynamics of interest (set by $\Omega V_0 \sim$ tens of MHz). These fast-oscillating terms time-average to zero over any gate duration and are dropped. This is the RWA.

After RWA, the drive Hamiltonian in the rotating frame is:

$$\hat{H}'_d = \frac{\Omega V_0 s(t)}{2}[(-I\cos\delta\omega t + Q\sin\delta\omega t)\hat{\sigma}_x + (I\sin\delta\omega t + Q\cos\delta\omega t)\hat{\sigma}_y]$$

In matrix form using complex exponential representation:

$$\hat{H}'_d = -\frac{1}{2}\frac{C_d}{C_\Sigma}\frac{\hbar}{\sqrt{2Z_0}}V_0 s(t)\begin{pmatrix} 0 & e^{i(\delta\omega t+\phi)} \\ e^{-i(\delta\omega t+\phi)} & 0 \end{pmatrix}$$

On-Resonance Simplification

Setting $\delta\omega = \omega_q - \omega_d = 0$ (drive exactly at qubit frequency), the Hamiltonian reduces to:

$$\hat{H}'_d = -\frac{\Omega V_0 s(t)}{2}(I\hat{\sigma}_x + Q\hat{\sigma}_y)$$

This is a static (non-oscillating) Hamiltonian in the rotating frame—exactly the form needed for universal qubit control:

- $\phi = 0$ (pure X drive): $I = 1, Q = 0 \to \hat{H}'_d = -\frac{\Omega V_0 s(t)}{2}\hat{\sigma}_x$ (rotation around X-axis)
- $\phi = \pi/2$ (pure Y drive): $I = 0, Q = 1 \to \hat{H}'_d = -\frac{\Omega V_0 s(t)}{2}\hat{\sigma}_y$ (rotation around Y-axis)
- Arbitrary ϕ: $I = \cos\phi, Q = \sin\phi$ (rotation around an axis at angle ϕ in the XY plane)

4.5 Rotation Angle and Gate Implementation

The Rotation Angle Integral

The total rotation angle accumulated by a drive pulse with envelope $s(t)$ and amplitude V_0 is:

$$\Theta(t) = -\Omega V_0 \int_0^t s(t')dt'$$

where $\Omega = \frac{C_d}{C_\Sigma}\sqrt{\frac{\hbar}{2Z_0}}$ is a fixed hardware parameter. The integral $\int s(t')dt'$ is the pulse area—determined by the waveform shape and duration. To implement a specific gate, one adjusts:

- V_0 (pulse amplitude): directly scales Θ
- $s(t)$ waveform (shape): common choices are rectangular, Gaussian, or DRAG-shaped envelopes
- t_{pulse} (duration): longer pulse at fixed amplitude gives larger Θ

A sequence of gates $(\Theta_k, \Theta_{k-1}, \ldots, \Theta_0)$ is compiled into a sequence of microwave pulses, each with the appropriate amplitude, duration, phase, and shape.

Pulse Shaping and DRAG

Rectangular pulses have broad frequency spectra that can off-resonantly drive the $|1\rangle \to |2\rangle$ transition (separated from $|0\rangle \to |1\rangle$ by the anharmonicity $\alpha \sim -200$ MHz). This causes leakage out of the qubit subspace.

With a Gaussian envelope $s(t)$ on the X channel, the DRAG correction adds:

$$s_Q(t) = -\frac{1}{\alpha}\frac{ds_I(t)}{dt}$$

This doubles the effective gate fidelity for typical transmon anharmonicities, enabling gate errors below 10^{-3}.

Hardware: I/Q Modulation Setup

In the lab, a microwave local oscillator (LO) at frequency ω_{LO} is mixed with two low-frequency signals from an Arbitrary Waveform Generator (AWG) at frequency ω_{AWG}:

$$\omega_d = \omega_{\text{LO}} + \omega_{\text{AWG}}$$

The AWG generates the $I(t)$ and $Q(t)$ envelopes. An IQ mixer produces the final drive signal at ω_d. By generating multiple AWG tones ($\omega_{\text{AWG1}}, \omega_{\text{AWG2}}, \ldots$) and mixing them with the same LO, one can simultaneously address multiple qubits or readout resonators at different frequencies—a key requirement for scaled-up processors.

4.6 The Virtual Z Gate

Z Rotations and Phase Shifting

A physical Z gate rotates the Bloch vector around the Z-axis by angle ϕ. In the lab frame, this would require either:

1. Detuning the drive from ω_q for a controlled time (allowing free precession to accumulate Z rotation), or
2. Applying a flux pulse to shift ω_q temporarily.

Both approaches introduce extra time and potential errors. There is, however, a much simpler and error-free alternative: the virtual Z gate.

Why It Works

Recall that the rotation axis in the XY plane is set by the phase ϕ of the drive ($I = \cos\phi, Q = \sin\phi$). In the rotating frame, applying a Z rotation by angle ϕ before an X gate is equivalent to relabeling the X and Y axes—i.e., shifting all subsequent phase references by ϕ. Mathematically:

$$Z(\phi) \cdot X(\theta) \cdot Z^\dagger(\phi) = R_{\text{rotated axis}}(\theta)$$

So instead of explicitly applying $Z(\phi)$, we update the I/Q components for all future pulses:

$$I' = I\cos\phi - Q\sin\phi, \quad Q' = I\sin\phi + Q\cos\phi$$

This is a rotation of the (I, Q) vector by angle ϕ on the complex plane—a trivial AWG software update.

Benefits and Scope

Virtual Z gates are now the standard method for implementing Z-axis rotations in superconducting quantum processors:

- **Zero gate time:** No additional pulse is required, improving circuit depth capacity
- **Zero control error:** The rotation is exact to numerical precision, unaffected by pulse distortions or calibration drift
- **No additional decoherence:** Since no physical interaction with the qubit occurs, T_1 and T_2 are not impacted
- **Composable:** Any sequence of Z rotations can be accumulated and folded into a single phase update

In practice, hardware compilers (e.g., IBM Qiskit Transpiler, Google Cirq) automatically convert all Z gates in a quantum circuit to virtual Z gates at the pulse scheduling stage, then propagate the accumulated phase through all subsequent I/Q pulses.